



Afynix Digital

Artificial Intelligence Internship

Duration: 6 Weeks

Level: Intermediate → Advanced

WEEK 1 – Regression Project

Project: Real Estate Pricing For a Startup

Helping Hints:

- Start by checking:
 - Missing values (isnull())
 - Data types (categorical vs numerical)
- For EDA:
 - Use correlation heatmap to find strong relationships
 - Try scatter plots (price vs area, rooms, etc.)
- Feature Engineering Ideas:
 - Convert categorical → numerical (encoding)
 - Create new features (e.g., price per sqft)
- Model Tips:
 - Train Linear Regression first → baseline
 - Then try Random Forest → compare performance
- Overfitting Check:
- Interpretation:
 - Which features have strongest influence?

WEEK 2 – Classification Project

Project: Customer Churn – Save the Business

Helping Hints:

- First check:
 - Class imbalance (value_counts())
- Handle imbalance:
 - Try oversampling or class_weight
- Feature Engineering:
 - Convert categorical columns properly
- Model Strategy:
 - Start simple → Logistic Regression
 - Then move to tree-based models
- Evaluation:
 - Don't rely only on accuracy

- o Look at confusion matrix carefully:
 - False negatives are dangerous (missed churners!)
- Business Thinking:
 - o Which customers are at risk?
 - o What action can retain them?

WEEK 3 – Unsupervised Learning

Project: Customer Segmentation for Marketing Team

Helping Hints:

- Before clustering:
 - o Scale your data (very important!)
- Choosing clusters:
 - o Use elbow method
- K-Means:
 - o Works well for round clusters
- Visualization:
 - o Use PCA if too many features
- Analysis:
 - o Don't just label clusters → describe them:
 - High spenders? Low engagement?
- Business Thinking:
 - o How would marketing target each group?

WEEK 4 – Computer Vision (OpenCV)

Project: Smart Attendance System

Helping Hints:

- Start with:
 - o Webcam capture using OpenCV
- Face Detection:
 - o Use Haar Cascade classifier
- For recognition:
 - o Store face encodings
- Attendance Logic:
 - o Avoid duplicate entries
 - o Add timestamp for each entry
- File Handling:
 - o Save CSV properly using pandas
- UI (Bonus):
 - o Start simple → Streamlit buttons
- Testing:
 - o Test in different lighting conditions

WEEK 5 – Natural Language Processing

Project: AI Content Analyzer

Helping Hints:

- Text Cleaning Steps:
 - Lowercase
 - Remove punctuation
 - Remove stopwords
- Vectorization:
 - Use TF-IDF (don't use raw text)
- Model Strategy:
 - Start with Naive Bayes (fast baseline)
 - Then Logistic Regression
- Evaluation:
 - Compare precision & recall
 - Don't trust only accuracy
- Error Analysis:
 - Look at wrong predictions manually
- Real-world Thinking:
 - Where can this be used? (social media, news filtering)

WEEK 6 – Completion & Submission Week

This week is for:

- Completing pending projects
- Preparing GitHub repositories (optional)
- Posting projects on LinkedIn
- Final submission form



Mandatory LinkedIn Posting Rule

All interns must:

- Post their projects on LinkedIn
- Mention: **@Afynix Digital**
- Add project screenshots or videos
- Add GitHub link (optional)
- Use internship hashtags

Failure to post = No completion certificate.